

## FINITE FIELDS

COMPILED BY DINU ȘERBĂNESCU

### 1. PROBLEMS

#### 1.1. Polynomials.

- (1) Let  $F$  be a finite field such that the polynomial  $X^2 - 5$  is irreducible in  $F[X]$ . Prove that the polynomial  $X^5 + a$  is reducible in  $F[X]$  for all  $a \in F$ .
- (2) Let  $F$  be a finite field. Prove that  $1 + 1 = 0$  if and only if for any non-constant polynomial  $f \in F[X]$ , the polynomial  $f(X^2)$  is reducible in  $F[X]$ .
- (3) Let  $K$  be a finite field,  $n \geq 2$  an integer,  $f \in K[X]$  an irreducible polynomial of degree  $n$ , and  $g$  the product of all the nonconstant polynomials in  $K[X]$  of degree at most  $n - 1$ . Prove that  $f$  divides  $g - 1$ .
- (4) Let  $F$  be a finite field with  $q \geq 4$  elements. Call two polynomials  $f$  and  $g$  in  $F[X]$  *neighbours* if they have the same degree and they differ by exactly one coefficient. Show that any polynomial of degree  $q - 1$  from  $F[X]$  has a neighbour from  $F[X]$  that is reducible in  $F[X]$ , and also has a neighbour with no roots in  $F$ .
- (5) Let  $K$  be a finite field with  $q$  elements,  $q \geq 3$ . Suppose  $M$  is the set of polynomials of degree  $q - 2$  in  $K[X]$  that have all nonzero and pairwise distinct coefficients. Determine the number of polynomials in  $M$  that have  $q - 2$  distinct roots in  $K$ .

## 1.2. NT flavored problems.

- (6) Determine all prime numbers  $p$  of the form  $5n + 1$  given that the numbers  $1^5, 2^5, \dots, n^5$  leaves pairwise distinct remainders when divided by  $p$ .
- (7) Consider the sequence  $(a_n)_{n \geq 1}$  defined by  $a_1 = 4$  and  $a_{n+1} = a_n^2 - 2$ ,  $n \geq 1$ . Suppose  $p$  is prime and let  $n = 2^p - 1$ . Show that if  $n$  divides  $a_{p-1}$ , then  $n$  is prime. Lucas-Lehmer primality test
- (8) We define the sequence  $(a_n)$  as follows:  $a_0 = 2$  and  $a_{k+1} = 2a_k^2 - 1$  for  $k \geq 0$ . Show that if a prime number  $p > 2$  divides  $a_n$ , then  $2^{n+3}$  divides  $p^2 - 1$ .
- (9) Let  $F$  be the field of  $p^2$  elements, where  $p$  is an odd prime. Suppose  $S$  is a set of  $(p^2 - 1)/2$  distinct nonzero elements of  $F$  with the property that for each  $a \neq 0$  in  $F$ , exactly one of  $a$  and  $-a$  is in  $S$ . Let  $N$  be the number of elements in the intersection  $S \cap \{2a : a \in S\}$ . Prove that  $N$  is even.
- (10) Let  $p$  be a odd prime number.
  - (a) Show that among  $p + 2$  distinct points in the plane with integer coordinates, one can choose 3 distinct points, say  $A, B, C$ , such that  $2 \text{ area}[ABC]$  is divisible by  $p$ .
  - (b) Show that it is possible to choose a set of  $p + 1$  distinct points with integral coordinates, such that  $p$  does not divide  $2 \text{ area}[ABC]$  whenever  $A, B, C$  are distinct elements of the set.



## 2. SOLUTIONS

- (1) Let  $F$  be a finite field such that the polynomial  $X^2 - 5$  is irreducible in  $F[X]$ . Prove that the polynomial  $X^5 + a$  is reducible in  $F[X]$  for all  $a \in F$ .

Let  $p$  be the characteristic of  $F$  and notice that  $p \neq 2$ . Clearly, 5 is not a quadratic residue mod  $p$ . By quadratic reciprocity,  $p$  is not a quadratic residue mod 5, and either  $p \equiv 2 \pmod{5}$  or  $p \equiv 3 \pmod{5}$ . Since a quadratic polynomial over  $F_p$  is irreducible in  $F$  (as all quadratic polynomials over  $F_p$  split in  $F_{p^2}$ ),  $F$  has order  $p^n$  for some odd  $n$ . Now  $5 \nmid p^n - 1$ , and  $x \mapsto x^5$  is bijective on  $F^*$ . Therefore,  $a$  has a fifth root  $b$  in  $F$  for every  $a$ , and  $x - b$  is a factor of  $x^5 - a$ .

- (2) Let  $F$  be a finite field. Prove that  $1+1=0$  if and only if for any non-constant polynomial  $f \in F[X]$ , the polynomial  $f(X^2)$  is reducible in  $F[X]$ .

Suppose  $1 + 1 = 0$ , hence  $F = F_{2^n}$  for some  $n$ . The function  $\phi : F \rightarrow F$  defined by  $\phi(x) = x^2$  is injective:  $\phi(x) = \phi(y) \implies x^2 = y^2 \implies (x-y)(x+y) = 0 \implies (x-y)^2 = 0 \implies x = y$ . As  $F$  is finite,  $\phi$  is also surjective, hence any element of  $F$  is square.

Suppose  $f = \sum_{k=0}^n a_k X^k$ , then there exists  $b_k \in K$  such that  $a_k = b_k^2$  for  $k = 0, 1, \dots, n$ . It follows that  $f(X^2) = \sum_{k=0}^n b_k^2 X^{2k} = (\sum_{k=0}^n b_k X^k)^2 = \sum_{i=0}^n \sum_{j=0}^n b_i b_j X^{i+j}$ , so  $f(X^2)$  is reducible.

Conversely, let  $a \in F$  and  $f = X^2 - a$ . Since  $g = f(X^2) = X^4 - a$  is reducible, then  $g$  has a root in  $F$ . So the function  $\phi$  is surjective and consequently injective. Since  $\phi(1) = \phi(-1) = 1$ , we have  $1 = -1$ , hence  $1 + 1 = 0$ .

- (3) Let  $K$  be a finite field,  $n \geq 2$  an integer,  $f \in K[X]$  an irreducible polynomial of degree  $n$ , and  $g$  the product of all the nonconstant polynomials in  $K[X]$  of degree at most  $n-1$ . Prove that  $f$  divides  $g-1$ .

In any finite field the product of all the nonzero elements is  $-1$ . To see this, we can just pair each element with its inverse, except for 1 and  $-1$ , which remain unpaired. Applying this fact in the field  $K[X]/(f)$  we obtain that the product of all the nonzero polynomials in  $K[X]$  of degree at most  $n-1$  is  $-1 \pmod{f}$ . On the other hand, the product of all the nonzero constant polynomials in  $K[X]$  is the constant polynomial whose value is the product of all the nonzero elements in  $K$ , that

is  $-1$ . Combining the last to lines we obtain that the product of all the nonconstant polynomials in  $K[X]$  is  $1 \pmod{f}$ .

- (4) Let  $F$  be a finite field with  $q \geq 4$  elements. Call two polynomials  $f$  and  $g$  in  $F[X]$  *neighbours* if they have the same degree and they differ by exactly one coefficient. Show that any polynomial of degree  $q-1$  from  $F[X]$  has a neighbour from  $F[X]$  that is reducible in  $F[X]$ , and also has a neighbour with no roots in  $F$ .

Let  $f \in F[X]$ ,  $\deg f = q-1$ . If  $f(0) = 0$ , we choose the neighbor  $g = f + X$ , and if  $f(0) = \alpha \neq 0$ , we choose the neighbor  $h = f - \alpha$ . As  $g(0) = 0 = h(0)$ , both  $g$  and  $h$  are reducible in  $F[X]$ . Recall that

$$\alpha_s = \begin{cases} 0, & \text{if } (q-1) \nmid s, \\ -1, & \text{if } (q-1) \mid s, s \geq 1. \end{cases}$$

Let  $f = \sum_{i=0}^{q-1} a_i X^i$ . We distinguish the following two cases:  
Case 1:  $a_0 = 0$ . Since

$$\tilde{f}(\alpha) = \sum_{\alpha \in K} \sum_{i=0}^{q-1} a_i \alpha^i = \sum_{i=0}^{q-1} a_i \sum_{\alpha \in K} \alpha^i = -a_{q-1} \neq 0,$$

it turns out that the associated polynomial function  $\tilde{f} : K \rightarrow K$  is not surjective — otherwise,  $\sum_{\alpha \in K} \tilde{f}(\alpha) = \sum_{\alpha \in K} \alpha = 0$ .

Therefore, there exists  $s \in K^* \setminus \text{Im } \tilde{f}$ , because  $0 \in \text{Im } \tilde{f}$ . In this case, the polynomial  $g = f - s$  is a neighbor of  $f$  and has no roots in  $K$ .

Case 2:  $a_0 \neq 0$ . If  $f = a_{q-1}X^{q-1} + a_0$ , then  $\text{Im } \tilde{f} = \{a_0, a_{q-1} + a_0\}$ , so there exists  $s \in K^* \setminus \text{Im } \tilde{f}$ , because  $K$  has at least four elements. The polynomial  $g = f - s$  is a neighbor of  $f$  and has no roots in  $K$ . If there exists  $i \in \{1, 2, \dots, q-2\}$  such that  $a_i \neq 0$ , consider the polynomial  $h = X^{q-i-1}f$ . Since

$$\tilde{h}(\alpha) = \sum_{\alpha \in K} \sum_{j=0}^{q-1} a_j \alpha^{q-i+j-1} = -a_i \neq 0,$$

it follows that the associated polynomial function  $\tilde{h} : K \rightarrow K$  is not surjective.

Therefore, there exists  $s \in K^* \setminus \text{Im } \tilde{h}$ , since  $0 = \tilde{h}(0) \in \text{Im } \tilde{h}$ . The polynomial  $g = f - sX^i$  is adjacent to  $f$  and has no roots in  $K$ :  $\tilde{g}(0) = f(0) \neq 0$ , and if there would exist  $\alpha \in K^*$  such



that  $\tilde{g}(\alpha) = 0$ , then  $s = \alpha^{q-1}s = \alpha^{q-i-1}(s\alpha^i) = \alpha^{q-i-1}(\tilde{f}(\alpha) - \tilde{g}(\alpha)) = \alpha^{q-i-1}\tilde{f}(\alpha) = \tilde{h}(\alpha) \in \text{Im}\tilde{h}$  — contradiction.

- (5) Let  $K$  be a finite field with  $q$  elements,  $q \geq 3$ . Suppose  $M$  is the set of polynomials of degree  $q-2$  in  $K[X]$  that have all nonzero and pairwise distinct coefficients. Determine the number of polynomials in  $M$  that have  $q-2$  distinct roots in  $K$ .

Since any polynomial  $g$  in  $M$  is associated in divisibility with a unique polynomial  $f$  in  $M$  such that  $f(0) = 1$ , we will count polynomials  $f = 1 + a_1X + \dots + a_{q-2}X^{q-2}$  with the required property. Let  $f$  be a polynomial, let  $x_1, \dots, x_{q-2}$  be its roots and consider the matrices

$$A = \begin{pmatrix} 1 & a_1 & a_2 & \dots & a_{q-3} & a_{q-2} \\ a_{q-2} & 1 & a_1 & \dots & a_{q-4} & a_{q-3} \\ a_{q-3} & a_{q-2} & 1 & \dots & a_{q-5} & a_{q-4} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_2 & a_3 & a_4 & \dots & 1 & a_1 \\ a_1 & a_2 & a_3 & \dots & a_{q-2} & 1 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 & 1 \\ y & x_1 & x_2 & \dots & x_{q-3} & x_{q-2} \\ y^2 & x_1^2 & x_2^2 & \dots & x_{q-3}^2 & x_{q-2}^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ y^{q-3} & x_1^{q-3} & x_2^{q-3} & \dots & x_{q-3}^{q-3} & x_{q-2}^{q-3} \\ y^{q-2} & x_1^{q-2} & x_2^{q-2} & \dots & x_{q-3}^{q-2} & x_{q-2}^{q-2} \end{pmatrix},$$

where  $K^* = \{y, x_1, x_2, \dots, x_{q-2}\}$ . As  $x_{q-1} = 1$ ,  $x \in K^*$ , one has

$$AB = \begin{pmatrix} f(y) & 0 & 0 & \dots & 0 & 0 \\ yf(y) & 0 & 0 & \dots & 0 & 0 \\ y^2f(y) & 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ y^{q-3}f(y) & 0 & 0 & \dots & 0 & 0 \\ y^{q-2}f(y) & 0 & 0 & \dots & 0 & 0 \end{pmatrix}.$$

In particular,

$$\begin{vmatrix} a_1 & a_2 \\ 1 & a_1 \end{vmatrix} = \begin{vmatrix} a_1 & a_3 \\ 1 & a_2 \end{vmatrix} = \dots = \begin{vmatrix} a_1 & a_{q-2} \\ 1 & a_{q-3} \end{vmatrix} = 0,$$

that is,  $a_2 = a_1^2, a_3 = a_1^3, \dots, a_{q-2} = a_1^{q-2}$ , so the coefficient set of  $f$  is  $\{1, a_1, a_1^2, \dots, a_1^{q-2}\}$ . Since this set is equal to  $K^*$ , it follows that  $a_1$  is a generator of the multiplicative group  $(K^*, \cdot)$  and  $f = 1 + a_1X + a_1^2X^2 + \dots + a_1^{q-2}X^{q-2}$ .

Conversely, any polynomial  $f = 1 + aX + a^2X^2 + \dots + a^{q-2}X^{q-2}$ , where  $a$  is a generator of the multiplicative group  $(K^*, \cdot)$  satisfy the claim, since  $f(a^{-1}X) = 1 + X + X^2 + \dots + X^{q-2} = \prod_{\alpha \in K^* \setminus \{1\}} (X - \alpha)$ , so  $f$  are the roots of  $aX$ , where  $\alpha \in K^* \setminus \{1\}$ .

Since  $K^*$  has  $\varphi(q-1)$  generators, it follows that the required number is  $(q-1)\varphi(q-1)$ .

- (6) Determine all prime numbers  $p$  of the form  $5n+1$  given that the numbers  $1^5, 2^5, \dots, n^5$  leaves pairwise distinct remainders when divided by  $p$ .

Consider the polynomial  $f = X^n - 1 \in F_p[X]$ . Since  $f(k^5) = k^{5n} - 1 = k^{p-1} - 1 = 0, \forall k = 1, \dots, n$  and  $1^5, 2^5, \dots, n^5 \in F_p^*$  are distinct, it follows that these are (all) the roots of the polynomial  $f$ . From Viète's (first) relation one has  $1^5 + 2^5 + \dots + n^5 = 0$ .

The identity  $1^5 + 2^5 + \dots + n^5 = \frac{1}{12}(n^2(n+1)^2(2n^2+2n-1))$  and  $(p, n) = (p, n+1) = 1$  imply  $p$  divides  $2n^2 + 2n - 1$ , so  $p = 5n+1 \mid 50n^2 + 50n - 25 = 2(5n+1)(5n+4) - 33$ . Therefore  $p$  divides 33, implying  $\boxed{p=11}$ , which satisfy the claim:  $11 = 5 \cdot 2 + 1$  and  $1^5 \neq 2^5 \equiv -1 \pmod{11}$ .

- (7) Consider the sequence  $(a_n)_{n \geq 1}$  defined by  $a_1 = 4$  and  $a_{n+1} = a_n^2 - 2, n \geq 1$ . Suppose  $p$  is prime and let  $n = 2^p - 1$ . Show that if  $n$  divides  $a_{p-1}$ , then  $n$  is prime. Lucas-Lehmer primality test

Let  $\omega = 2 + \sqrt{3}, \bar{\omega} = 2 - \sqrt{3}$ . Obviously  $\omega\bar{\omega} = 1$ . By induction we obtain  $a_m = \omega^{2^m-1} + \bar{\omega}^{2^m-1}$ . As  $n$  divides  $a_{p-1}$ , the result is  $\omega^{2^{p-2}} + \bar{\omega}^{2^{p-2}} \equiv 0 \pmod{n}$ . Assume that  $n$  is not prime and let  $q$  be an odd prime divisor of  $n$ , with  $q^2 \leq n$ . As  $\omega^{2^{p-2}} + \bar{\omega}^{2^{p-2}} \equiv 0 \pmod{q}$ , using  $\omega\bar{\omega} = 1$  we successively obtain  $\omega^{2^{p-2}} \equiv -\bar{\omega}^{2^{p-2}} \equiv 0 \pmod{q} \implies \omega^{2^{p-1}} \equiv -1 \pmod{q} \implies \omega^{2^p} \equiv 1 \pmod{q}$ , in other words  $\omega$  has order  $2^p$  modulo  $q$ . On the other hand, the field  $F_q(\sqrt{3})$  has  $q$  or  $q^2$  elements, depending on whether  $\sqrt{3}$  is or is not a quadratic remainder modulo  $q$ .

In both situations we have  $\omega^{q^2-1} \equiv 1 \pmod{q}$ . Hence  $2^p \mid q^2 - 1$ , so  $2^p < q^2 < n = 2^p - 1$ , contradiction.



- (8) We define the sequence  $(a_n)$  as follows:  $a_0 = 2$  and  $a_{k+1} = 2a_k^2 - 1$  for  $k \geq 0$ . Show that if a prime number  $p > 2$  divides  $a_n$ , then  $2^{n+3}$  divides  $p^2 - 1$ .

Define the sequence  $b_n$  such that  $a_n = \frac{1}{2} \left( b_n + \frac{1}{b_n} \right)$ . Then the equation reads  $b_{n+1} = b_n^2$  with  $b_0 = 2 + \sqrt{3}$ . Using this we have  $a_n = \frac{1}{2} \left( (2 + \sqrt{3})^{2^n} + (2 - \sqrt{3})^{2^n} \right)$ . We will work in  $F_p(\sqrt{3})$  which may be  $F_p$  if 3 is a quadratic residue mod  $p$  and  $F_{p^2}$  otherwise. Note that if  $\theta = 2 + \sqrt{3}$ , we write  $a_n = 0 \iff \theta^{2^n} + \theta^{-2^n} = 0 \iff \theta^{2^{n+2}} = 1$ . Since  $F_p(\sqrt{3}) - \{0\}$  is cyclic, we have  $2^{n+2} \mid p^2 - 1$ . It boils down to proving that  $\sqrt{\theta}$  exists in  $F_p[\sqrt{3}]$ .

Note that  $\theta = 2 + \sqrt{3} = \frac{1}{2} (1 + \sqrt{3})^2$ . So we need that  $\left(\frac{2}{p}\right) = 1$  or, equivalently,  $8 \mid p^2 - 1$ . However, we have  $8 \mid 2^{n+2} \mid p^2 - 1$ , so we are done.

- (9) Let  $F$  be the field of  $p^2$  elements, where  $p$  is an odd prime. Suppose  $S$  is a set of  $(p^2 - 1)/2$  distinct nonzero elements of  $F$  with the property that for each  $a \neq 0$  in  $F$ , exactly one of  $a$  and  $-a$  is in  $S$ . Let  $N$  be the number of elements in the intersection  $S \cap \{2a : a \in S\}$ . Prove that  $N$  is even.

For  $a \in S$ , there is a unique way to write  $2a = \varepsilon_a s_a$  where  $\varepsilon_a = \pm 1$  and  $s_a \in S$ . Then  $S \cap 2S = \{a \in S : \varepsilon_a = 1\}$ , so

$$\prod_{a \in S} \varepsilon_a = (-1)^{\#S-N} = (-1)^N,$$

since  $\#S = (p-1) \cdot (p+1)/2$  is even. In  $F$ , we have

$$\prod_{a \in S} 2^{(p^2-1)/2} a = \prod_{a \in S} \varepsilon_a s_a = (-1)^N \prod_{a \in S} a,$$

so  $(-1)^N = 2^{(p^2-1)/2} = (2^{p-1})^{(p+1)/2} = 1^{(p+1)/2} = 1$ , by Fermat's Little Theorem. Hence  $N$  is even.

- (10) Let  $p$  be a odd prime number.

- (a) Show that among  $p+2$  distinct points in the plane with integer coordinates, one can choose 3 distinct points, say  $A, B, C$ , such that  $2 \text{area}[ABC]$  is divisible by  $p$ .
- (b) Show that it is possible to choose a set of  $p+1$  distinct points with integral coordinates, such that  $p$  does not divide  $2 \text{area}[ABC]$  whenever  $A, B, C$  are distinct elements of the set.

(a) Suppose that  $P_i = (x_i, y_i)$ ,  $i = 0, 1, 2, \dots, p+1$  are points, no 3 on a line such that  $2 \text{area}(\triangle P_i P_j P_k)$  is not divisible by  $p$  for all  $i, j, k$  with  $1 \leq i < j < k \leq p+2$ . After translation, we

may assume that  $P_0 = (0, 0)$ . We have

$$2\text{Area}(\triangle P_i P_j P_k) = \pm \det \begin{pmatrix} 1 & 1 & 1 \\ x_i & x_j & x_k \\ y_i & y_j & y_k \end{pmatrix}.$$

Let  $F_p = \mathbb{Z}/p\mathbb{Z}$  be the integers modulo  $p$  and define  $P_i = (x_i, y_i) \in F_p^2$  and  $x_i, y_i \in F_p$  are the residue classes modulo  $p$  of  $x_i$  and  $y_i$  respectively for all  $i$ . Modulo  $p$  we have

$$0 \neq 2\text{Area}(\triangle P_i P_j P_k) \pmod{p} = \pm \det \begin{pmatrix} 1 & 1 & 1 \\ x_i & x_j & x_k \\ y_i & y_j & y_k \end{pmatrix}$$

for all  $i, j, k$ . Let  $T = \{P_0, \dots, P_{p+1}\} \subseteq F_p^2$ . The nonvanishing of (1) means that no 3 points of  $T$  are colinear. In  $F_p^2$ , there are  $p+1$  rays through the origin  $P_0 = (0, 0)$ . At most 1 of the points in  $T \setminus \{P_0\}$  can lie on each of these rays. Therefore, exactly 1 of these points lies on each ray through the origin. So each ray through  $P_0$  contains two elements of  $T$ . Similarly, every line through  $P_i$  contains 2 points from  $T$  for  $i = 1, 2, \dots, p+1$ . This shows that every line in  $F_p^2$  contains 0 or 2 points from  $T$ . Consider the lines  $x = 0, 1, \dots, p-1$ . Each of these lines contains an even number of elements of  $T$ . Since the plane is a disjoint union of these lines,  $T$  has an even number of elements. But  $T$  has  $p+2$  elements and  $p$  is odd. Contradiction.

(b) Let  $F_{p^2}$  be the field with  $p^2$  elements. Then  $F_{p^2}$  has a primitive  $(p+1)$ -th root of unity  $\alpha$ . We can identify  $F_{p^2}$  with  $F_p^2$  as an  $\mathbb{F}_p$ -vector space. We claim that among  $1, \alpha, \alpha^2, \dots, \alpha^p$  there are no three lying on a line. Note that  $\beta, \gamma, \delta$  lie on a line if and only if  $\frac{\beta-\delta}{\gamma-\delta} \in F_p$ . Suppose that  $\alpha^i, \alpha^j, \alpha^k$  lie on a line. Then  $\frac{\alpha^k - \alpha^i}{\alpha^j - \alpha^i} = \beta \in F_p$ . Applying Frobenius and noting that  $\alpha^p = \alpha^{-1}$  and  $\beta^p = \beta$  we get  $\frac{\alpha^{p-k} - \alpha^{-i}}{\alpha^{p-j} - \alpha^{-i}} = \beta$ . Dividing the first equation by the second yields  $\alpha^{k-j} = 1$ . This is a contradiction because  $1 \leq k-j < p+1$ . We can lift  $\alpha_1, \dots, \alpha_{p+1}$  to points in  $\mathbb{Z}^2$  with the desired property.